

OptiMax-i

PICAN

- On the EPDAF print, there is a PICAN node. This node connects the print to the different PICAN devices.
- The EPDAF print has 2 jumpers (EPDAF17 and EPDAF18). These jumpers connect the 2 terminations.
- The EPDAF print has measuring points where the PICAN signals can be measured.
- For more information, see: PICAN bus [86]

Replacement of the print



When replacing the EPDAF print, the position of the terminations (jumpers) must be checked, see: Termination on the EPDAF [88].

TRIAX print

Name

TRI AXes

Version

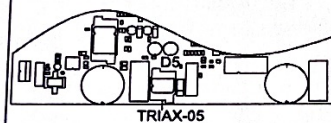
Version	Machine version	This version controls...	Can be replaced by ...
TRIAX/D-2: BE303188	• single beam, without Optileno.	max. 2 servomotors: • ETU; • ELO1 (RHS).	• TRIAX/D-2; • TRIAX-2 => some connectors are not used.
TRIAX-2: BE302460	• single beam; • double beam; • fancy let-off on the machine.	max. 3 servomotors: • ETU; • ELO1 (RHS); • ELO2 (LHS); • ELO-FA	TRIAX-2

Safety

Before working on the print:

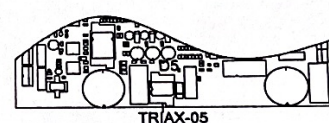
1. Set main switch QM to position -0-.
2. Wait for the green LED D5 to fade completely, because the capacitors discharge only slowly.

TRIAX/D-2



TRIAX-05

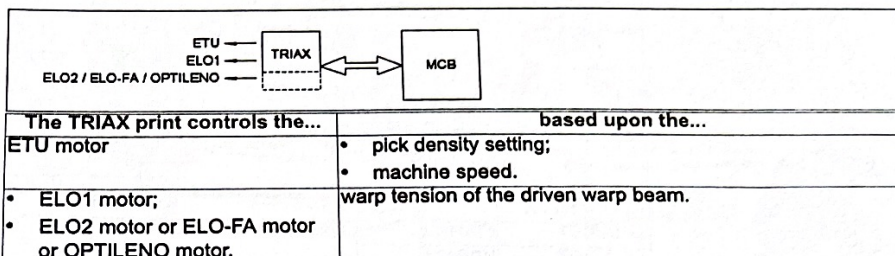
TRIAX-2



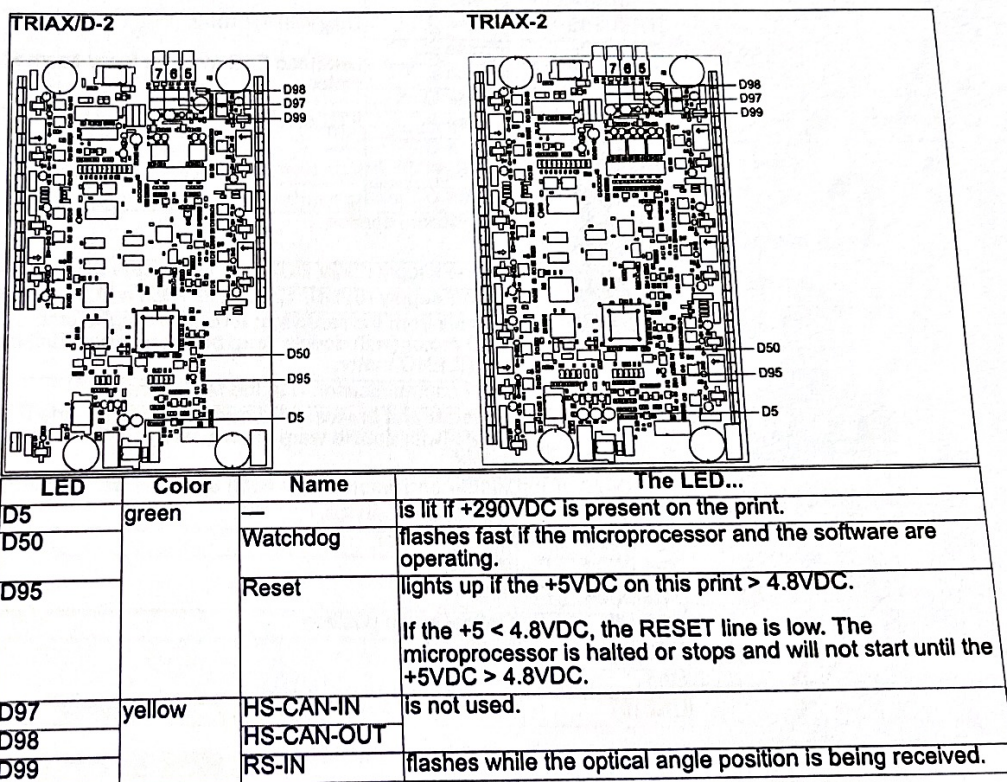
TRIAX-05

- This print contains a microprocessor using its own software. This software is checked upon:
 - power-on;
 - each change of the machine software.
- If the check indicates that the software is not compatible with the MCB print software, new software is automatically loaded into the TRIAX print.
- This is an air-cooled print. The cooling system of this print can be severely disrupted by dirt such as dust (dressing). As a consequence, to ensure the proper operation of the TRIAX print it is very important that the air circulation on and around the cooling plate of the print is kept optimal. The heatsink of the print should be blown clean regularly (with every gaiting).

Function



Signals

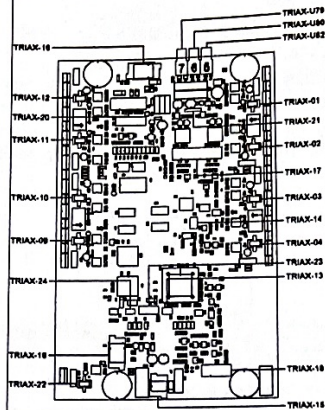


Fusing

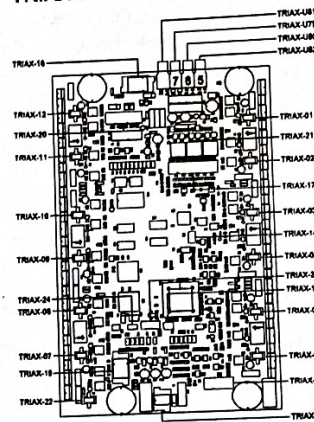
There are no fuses on the print.

Connections

TRIAX/D-2

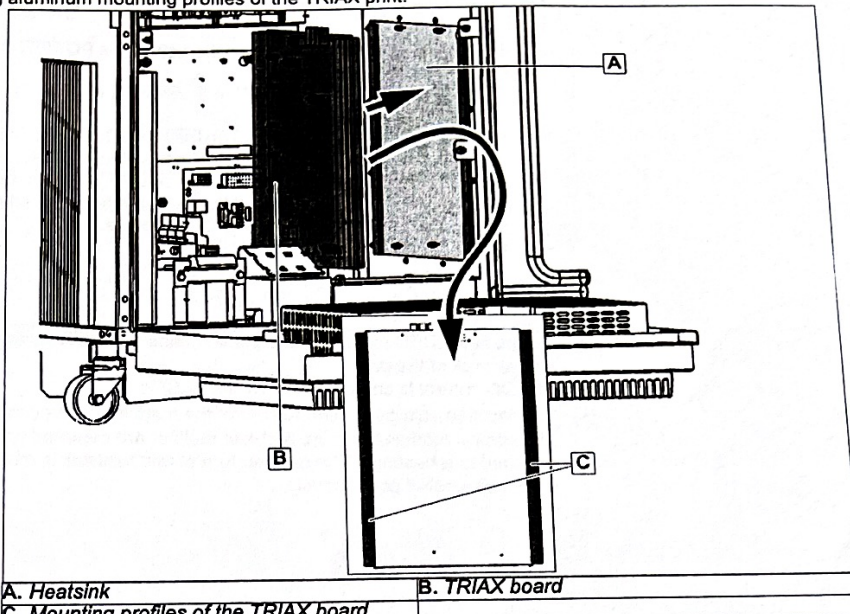


TRIAX-2

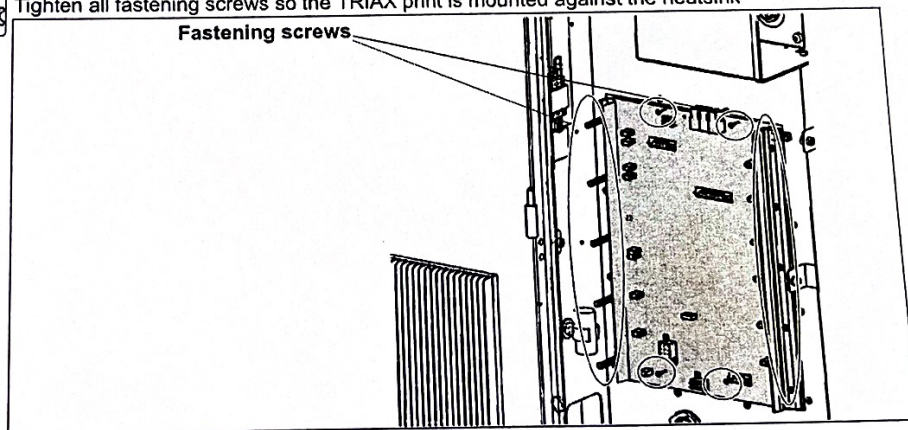


Connector	Connections	
TRIAX-01	Phase A	Right-hand ELO motor.
TRIAX-02	Phase C	
TRIAX-03	Phase B	
TRIAX-04	Phase D	
TRIAX-05	Phase A	Only with TRIAX-2.
TRIAX-06	Phase C	Left-hand ELO motor or fancy beam ELO motor or OPTILENO motor.
TRIAX-07	Phase B	
TRIAX-08	Phase D	ETU motor.
TRIAX-09	Phase A	
TRIAX-10	Phase C	
TRIAX-11	Phase B	
TRIAX-12	Phase D	
TRIAX-13	For Picanol service.	
TRIAX-14		
TRIAX-15	Power supply (290V DC) coming from the EPDAF print.	
TRIAX-16	Power supply (5VDC; 12VDC; 26VDC) and DD-2 coming from the EPDAF print.	
TRIAX-17	Signals from the resolvers: ETU motor, right-hand side ELO motor, left-hand side ELO motor (with double warp beam) or fancy ELO motor (with fancy beam) or OPTILENO motor.	
TRIAX-18	CAN communication with the MCB print.	
TRIAX-19	Bimetals and brakes: ETU motor, right-hand side ELO motor, left-hand side ELO motor (with double warp beam), fancy ELO motor (with fancy beam) or OPTILENO motor.	
TRIAX-20	Right-hand waste cutter (with electrical waste cutters).	
TRIAX-21	For Picanol service.	
TRIAX-22	Dump resistor.	
TRIAX-23	Not used.	
TRIAX-24		
U79 (7)	Optical angle position.	
U80 (6)	Not used.	
U81		
U82 (5)		

When replacing the TRIAX print, apply some cooling paste between the heatsink and the 2 aluminum mounting profiles of the TRIAX print.



Tighten all fastening screws so the TRIAX print is mounted against the heatsink



NG_LDEC

Name

New Generation Loom Drive Electronic Converter

Version

NG_LDEC: BE232105

Safety

 The NG_LDEC module contains $\pm 600V$ DC.

Before working on the NG_LDEC module:

1. Switch off the power supply of the main control box.
2. Wait 3 minutes before opening the NG_LDEC (the capacitors discharge only slowly).

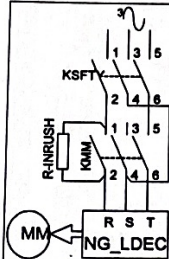
General

- The 3-phase mains voltage is used as power supply for controlling the motor phases. The mains voltage is usable in the 380V AC to 460V AC range.



If the weaving mill voltage is not in this range, a transformer is used.

- Upon activation of the machine and after pushing the POWER ON button, the following steps are performed:

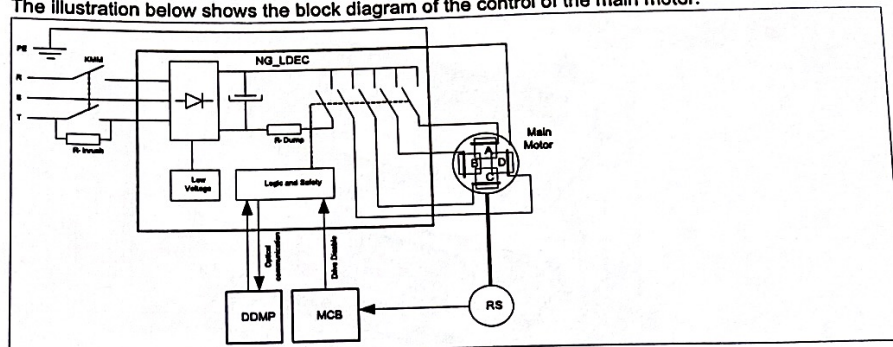


- Only 1 of the 3 phases is connected to the NG_LDEC module via the INRUSH print.
- The ELCO on the NG_LDEC module is charged slowly and the tension on the ELCO is continuously measured.
- If the tension becomes higher than the value set by the software, the K-MM motor switch can be switched on so all 3 phases come directly to the NG_LDEC module.

- On the NG_LDEC module, the 3-phase mains voltage is rectified by a 3-phase rectifier located on the back of the print. The DC current is smoothed out by the ELCOs.
- The rectified voltage is used to control the main motor via control modules. The control modules and the 3-phase rectifier are mounted on the NG_LDEC module and attached to a heatsink. The temperature of this heatsink is monitored continuously. This heatsink is also air-cooled continuously.

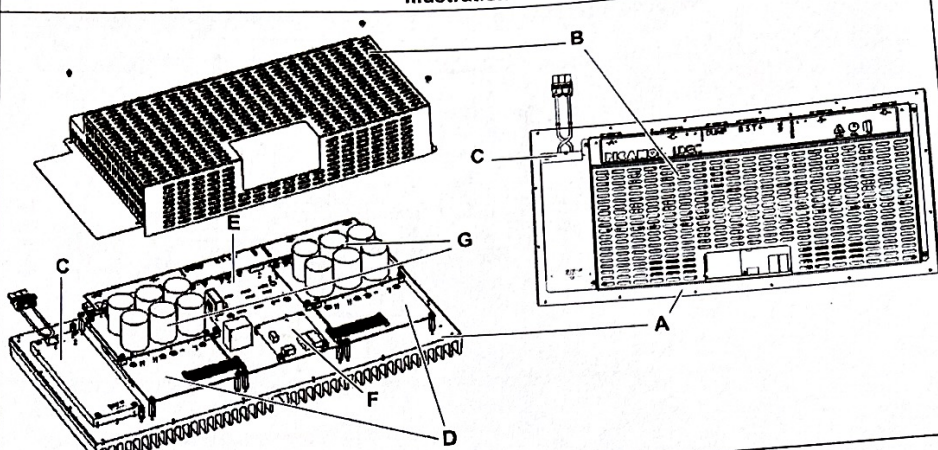
The NG_LDEC controls, checks and protects the control of the main motor. The NG_LDEC receives the required signals from MCB and DDMP.

The illustration below shows the block diagram of the control of the main motor:



Composition of the
NG_LDEC

The NG_LDEC consists of various components and prints. These are:

Illustration	
	
Component	Name
A	Heatsink
B	Cover
C	Dump resistor
D	NG_LDEC > OBMM110 print [52]
E	NG_LDEC > RBMM110 print [56]
F	NG_LDEC > CBMM110 print [60]
G	NG_LDEC > EBMM1020 print [62]

Protection

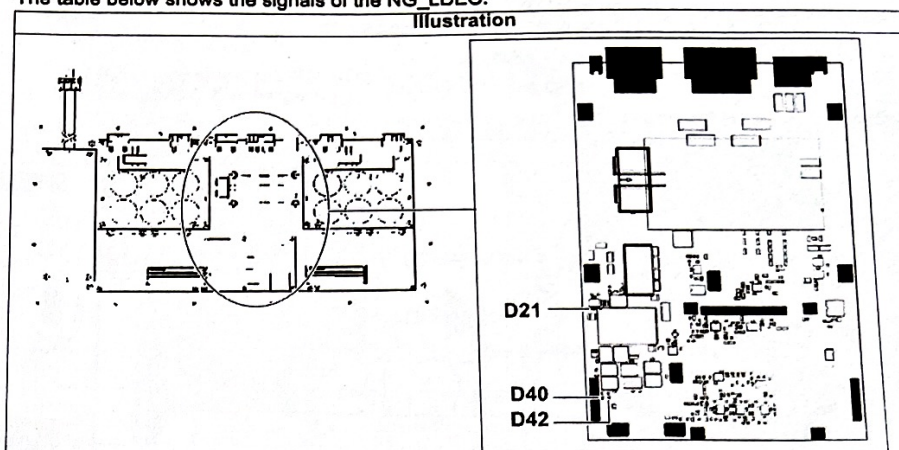
The NG_LDEC disengages the main motor or prevents it from starting if one of the following conditions is exceeded:

- Overvoltage motor phases: 250A
- Differential: $\pm 2A$;
- Overvoltage DC: 796V;
- Undervoltage DC: 370V;
- Temperature of the modules: 80°C;
- Low voltage check;
- Desaturation protection.

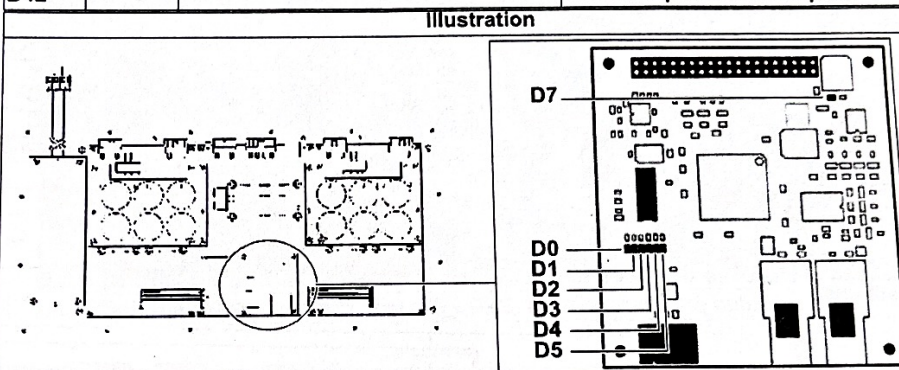


If an error occurs, an error message will appear on screen and LEDs will light up on the NG_LDEC. See also: Error detection [48].

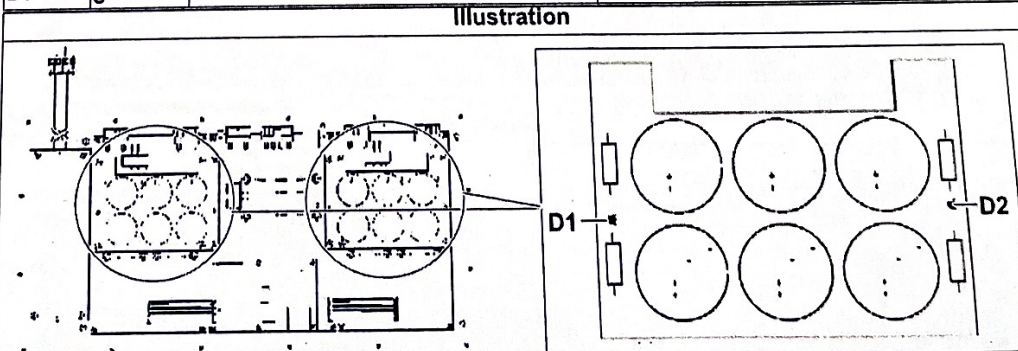
The table below shows the signals of the NG_LDEC:



LED ...	color ...	on the ...	lights up ...
D21	green	NG_LDEC > RBMM110 print [56]	if VDCL is present on the print.
D40			if 26VDC is present on the print.
D42			if 5VDC is present on the print.



LED ...	color ...	on the ...	lights up ...
D0	orange	NG_LDEC > CBMM110 print [60]	at error detection of the NG_LDEC. See Error detection [48]
D1			
D2			
D3			
D4			
D5			if the drive is activated.
D7	green		if 3,3V is present on the print.



LED ...	color ...	on the ...	lights up ...
D1	red	NG_LDEC > EBMM1020 print [62] (EBMMac)	if the ELCOs are under tension.
D2			
D1		NG_LDEC > EBMM1020 print [62] (EBMMbd)	
D2			

Error detection

On the NG_LDEC, there are 6 LEDs that, when lit, refer to a specific error.
The table below shows the explanation of the possible lit LED combinations.



1: LED lights up.

0: LED does not light up.

Illustration

D4	D3	D2	D1	D0	Error (explanation error message LEDs)	Error message on screen
0	0	0	0	1	Drive Enable External	FDD and/or dtisc
0	0	0	1	0	Optocoupler supply	DTISC
0	0	0	1	1	Fault feedback	DTISC
0	0	1	0	0	Low voltages OK	VLSOK and/or PBEsw
0	0	1	0	1	Motor current	oc_sw
0	0	1	1	0	Differential current	DTISC
0	0	1	1	1	Undervoltage safety	DTISC
0	1	0	0	0	Drive Enable External	PBEsw and/or DTISC
D5						
1					Drive is activated	
0					Drive is deactivated	

Connections

Illustration	
Connector	Connections
-A+	phase A
+C-	phase C
DTTHDUMP	is not used
DUMP	connection dump resistor
RST	3-phase input voltage and mass
VDCL	is not used
DTTHMM	
-B+	phase B
+D-	phase D
DD	Drive Disable
U3 (2)	optical connection to DDMP-U4 (2) (receiver)
U4 (1)	optical connection to DDMP-U2 (1) (transmitter)

Removing the NG_LDEC

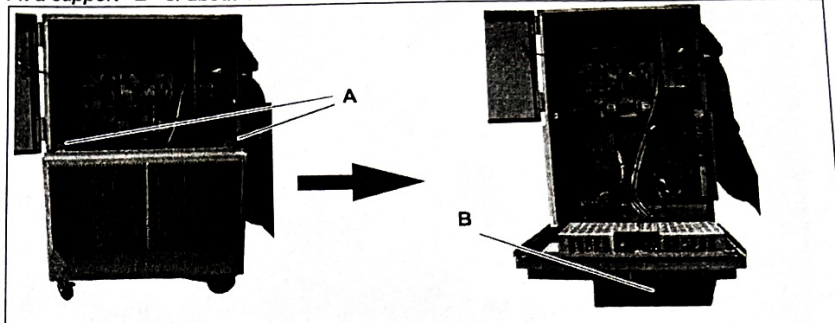
1. Switch off the main power supply.
2. Open the power box and tip the NG_LDEC over by removing the 2 fastening screws "A" and lower the panel slowly.



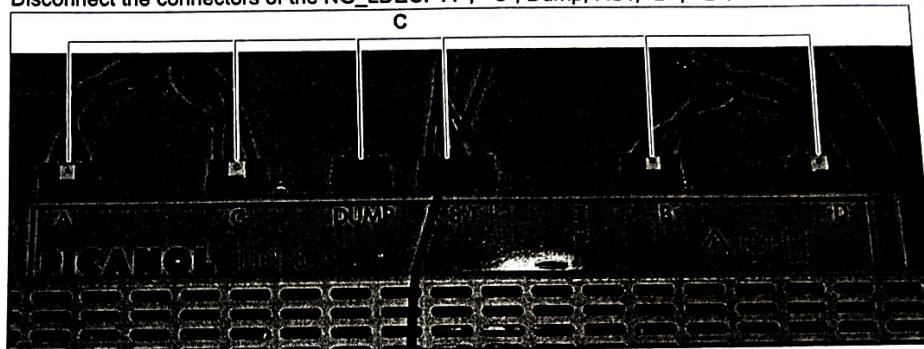
Before opening the NG_LDEC panel, wait 3 minutes until the capacitors have completely discharged.



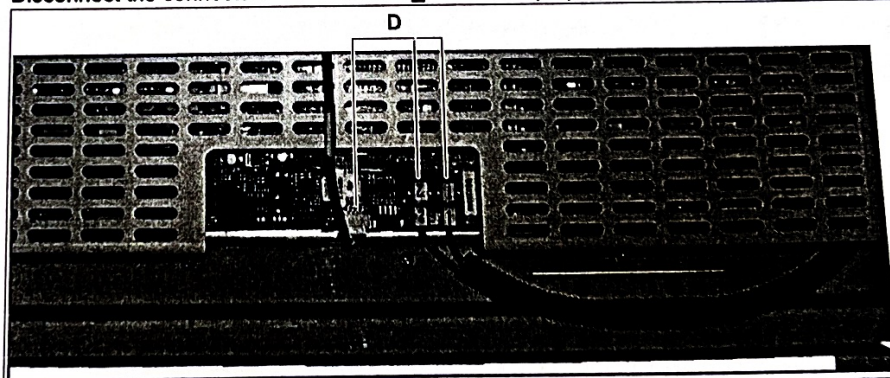
Fit a support "B" of about 4 cm underneath the NG_LDEC panel, so the panel is horizontal.



3. Disconnect the connectors of the NG_LDEC: -A+; +C-; Dump; RST; -B+; +D-.



4. Disconnect the connectors "D" of the NG_LDEC: DD (P2); optical 2; optical 1.



5. Disconnect the 5 earth and shielding connections "E".

